



Coinbase: base contracts

Security Review

Cantina Managed review by:

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1 Introduction

1.1 About Cantina

Cantina is a security services marketplace that connects top security researchers and solutions with clients. Learn more at cantina.xyz

1.2 Disclaimer

Cantina Managed provides a detailed evaluation of the security posture of the code at a particular moment based on the information available at the time of the review. While Cantina Managed endeavors to identify and disclose all potential security issues, it cannot guarantee that every vulnerability will be detected or that the code will be entirely secure against all possible attacks. The assessment is conducted based on the specific commit and version of the code provided. Any subsequent modifications to the code may introduce new vulnerabilities that were absent during the initial review. Therefore, any changes made to the code require a new security review to ensure that the code remains secure. Please be advised that the Cantina Managed security review is not a replacement for continuous security measures such as penetration testing, vulnerability scanning, and regular code reviews.

1.3 Risk assessment

Severity level	Impact: High	Impact: Medium	Impact: Low
Likelihood: high	Critical	High	Medium
Likelihood: medium	High	Medium	Low
Likelihood: low	Medium	Low	Low

1.3.1 Severity Classification

The severity of security issues found during the security review is categorized based on the above table. Critical findings have a high likelihood of being exploited and must be addressed immediately. High findings are almost certain to occur, easy to perform, or not easy but highly incentivized thus must be fixed as soon as possible.

Medium findings are conditionally possible or incentivized but are still relatively likely to occur and should be addressed. Low findings are a rare combination of circumstances to exploit, or offer little to no incentive to exploit but are recommended to be addressed.

Lastly, some findings might represent objective improvements that should be addressed but do not impact the project's overall security (Gas and Informational findings).

2 Security Review Summary

Base is a secure, low-cost, builder-friendly Ethereum L2 built to bring the next billion users onchain.

From Aug 28th to Aug 31st the Cantina team conducted a review of [contracts](#). The review covered the changes in four pull requests: [#250](#), [#378](#), [#399](#), and [#405](#). During the review the team raised no issues.

2.1 Scope

The following files were affected by the above referenced PRs and were included in the scope of this security review::

```
src
├── L1
│   ├── ETHLockbox.sol (Deleted)
│   ├── OptimismPortal2.sol
│   ├── SystemConfig.sol
│   └── proofs
│       ├── DisputeGameFactory.sol
│       └── tee
│           └── TEEProverRegistry.sol
└── libraries
    └── Features.sol
```

3 Overview

The PRs of this codebase included three big ideas:

1. To update the DisputeGameFactory to deploy games using CREATE2 instead of CREATE
2. To remove the ETHLockbox from the ETH bridging process in the OptimismPortal2 contracts
3. Allow the on-chain NitroValidator to validate AWS Nitro attestations for Nitro signer registration

These are some big changes in theory, however the existing processes and infrastructure were designed to allow these changes to be made simply and atomically with very few complications. Simple execution branches were removed or an external call added to make these larger changes.

4 Additional Comments

Overall we were impressed with the codebase. Design decisions made well in advance made the changes here simple and smooth to effect and review. We commend Coinbase for their high code quality and thank them for being respondent during the engagement.